AICS201

AI Ethics, Governance and Legal Compliance Assessment

Assessment of the fictional SecureFinance Africa AI-powered financial decision system

Project	AICS201_AI_Ethics_Governance_Assessment
Assessment focus	Fairness, explainability, risk governance, legal compliance and mitigation
Evidence basis	Synthetic dataset and project analysis outputs
Report date	11 November 2026
Repository	github.com/aicdf2616779/AICS201_AI_Ethics_Governance_Assessment

Prepared as a project assessment report. All dataset records and model outcomes discussed here are synthetic and should not be interpreted as evidence about real applicants, a real lender, or a deployed financial system.

CONFIDENTIALITY / USE NOTE

This report is intended to document the academic project implementation and its analytical findings. It is not legal advice, a production model validation, or an authorisation to make real credit decisions.

Executive Summary

This project evaluates ethical, governance and legal-compliance considerations for a fictional AI-enabled financial decision system, SecureFinance Africa. The work combines a synthetic dataset, group fairness measurements, surrogate-model explainability, bias-mitigation experiments, intersectional analysis, an AI risk register and a preliminary compliance assessment.

The initial fairness audit identified measurable differences in model outcomes across gender and region. In the initial evaluation, the female selection rate was 0.3561 compared with 0.4803 for males, a demographic-parity difference of 0.1242. The rural selection rate was 0.3755 compared with 0.4420 for urban applicants, a difference of 0.0665. True-positive-rate differences were 0.2511 for gender and 0.1639 for region. These figures indicate disparities in this synthetic experiment; they do not, by themselves, establish unlawful discrimination or explain its cause.

The explainability experiment used a surrogate model with test accuracy of 0.9033. The largest mean absolute SHAP contributions were debt-to-income ratio (0.1626), monthly income (0.1332), credit-history months (0.1053) and missed payments (0.0887). These values describe the surrogate model's behaviour and are not causal effects or proof that the original system uses a feature in a particular way.

Mitigation experiments improved some measured group gaps but exposed trade-offs. The gender-constrained model reduced the gender demographic-parity difference to 0.0162, while its regional demographic-parity difference increased to 0.1383. The region-constrained model achieved a gender difference of 0.0418 and a regional difference of 0.1271. Both mitigation models achieved 0.94 agreement with the synthetic reference labels; this is not real-world lending accuracy.

The project also documented ten AI risks, prepared a governance framework aligned to the NIST AI Risk Management Framework functions (GOVERN, MAP, MEASURE and MANAGE), and produced a compliance matrix covering the Nigeria Data Protection Act 2023, the EU AI Act, UNESCO and OECD materials. The compliance matrix is preliminary and requires provision-by-provision legal review and an applicability assessment before any real deployment.

Overall conclusion: the project demonstrates a repeatable assessment workflow and highlights areas requiring further investigation, controls and human oversight. No production-readiness or legal-compliance certification is claimed.

Key project outputs

Area	Output / result
Dataset	1,200 synthetic records, plus header
Fairness audit	Group selection rates, TPR, demographic-parity and equal-opportunity comparisons
Explainability	Surrogate model evaluation and SHAP feature-importance outputs
Risk governance	10-row AI risk register and governance framework
Legal assessment	13-entry compliance matrix and preliminary legal assessment
Mitigation	Gender- and region-constrained experiments; comparison CSV and chart
Testing	Six automated project-validation tests passed

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1. Introduction and Scope

The project examines an illustrative AI-powered financial decision environment referred to as SecureFinance Africa. The purpose is to assess how an AI system could be evaluated for fairness, transparency, accountability, governance and legal-compliance considerations. The assessment is a technical and documentary exercise using synthetic data; it does not audit a real organisation or a live credit-scoring platform.

Objectives

• Generate and document a synthetic financial dataset.
• Measure outcome differences across gender and region groups.
• Examine model behaviour using SHAP-based feature attribution on a surrogate model.
• Experiment with fairness-constrained mitigation and compare trade-offs.
• Assess intersectional group outcomes.
• Document AI risks, governance controls and preliminary legal obligations.
• Validate project artefacts using automated tests.

Scope boundary. The work does not include access to real customer records, a live production model, a formal legal opinion, independent model validation, security penetration testing, or a deployment approval. The fictional scenario is used to structure the assessment.

2. Methodology and Project Architecture

The repository is organised into data, scripts, fairness analysis, explainability, risk assessment, legal compliance, documentation, outputs and tests. The analysis is implemented in Python. The reported results are drawn from the project outputs and execution summaries established during the project workflow.

Project area	Main artefacts
Data and generation	data/synthetic/securefinance_synthetic_data.csv; scripts/generate_synthetic_data.py; docs/DATA_DICTIONARY.md
Fairness	fairness_analysis/fairness_audit.py
Mitigation	fairness_analysis/bias_mitigation.py
Intersectional analysis	fairness_analysis/intersectional_audit.py
Explainability	explainability/shap_analysis.py
Governance	risk_assessment/ai_risk_register.csv; docs/AI_GOVERNANCE_FRAMEWORK.md
Legal assessment	legal_compliance/legal_compliance_assessment.md; legal_compliance/compliance_matrix.csv
Validation	tests/test_project_validation.py

3. Dataset and Analytical Approach

The project uses a synthetic dataset containing 1,200 records. The data are simulated for assessment purposes and must not be treated as a representative sample of the Nigerian population or of actual credit applicants. The dataset includes demographic group attributes and financial/credit-related variables used by the project analyses.

The fairness audit compared selection rates and true-positive rates by gender and region. Demographic-parity difference is reported as the absolute difference between group selection rates; the selection-rate ratio is the lower group rate divided by the higher group rate. Equal-opportunity difference is the absolute difference between group true-positive rates. These metrics are diagnostic summaries and should be interpreted together with sample sizes, label quality, model context and the consequences of errors.

For mitigation, the project used a 75/25 train/test split with random_state 42 and Fairlearn's ExponentiatedGradient approach with DemographicParity constraints. The gender- and region-constrained models were evaluated separately. The comparison used a synthetic neutral_approved reference label. Consequently, reported accuracy is agreement with that synthetic reference, not predictive validity against real repayment outcomes.

4. Initial Fairness Audit

The initial audit reported the following group metrics. Selection rate represents the proportion of each group receiving a positive model outcome. TPR represents the proportion of positive reference cases correctly receiving a positive model outcome.

Group	Selection rate	TPR	Sample size
Female	0.3561	0.6887	615
Male	0.4803	0.9398	585
Rural	0.3755	0.7107	458
Urban	0.4420	0.8747	742

Comparison	DP difference	Selection-rate ratio	Equal-opportunity difference
Gender	0.1242	0.7413	0.2511
Region	0.0665	0.8496	0.1639

The largest initial disparity in this set of metrics is between gender groups: a selection-rate difference of approximately 12.42 percentage points and a TPR difference of approximately 25.11 percentage points. The regional comparison also shows differences, though smaller on selection rate. The confusion matrix was [[583, 0], [117, 500]], with reported accuracy 0.9025.

Interpretation and caution

The metrics flag areas for investigation rather than provide a legal finding. A disparity can arise from multiple factors, including data construction, label definitions, sampling, feature relationships, model design and decision thresholds. The synthetic dataset and reference labels limit any external inference. The next stage in a real setting would require data provenance review, label-quality testing, error analysis, confidence intervals and consultation with affected stakeholders.

5. Explainability Analysis

The explainability workflow evaluated a surrogate model and calculated SHAP-based feature contributions. The surrogate achieved test accuracy of 0.9033. The mean absolute SHAP values below rank features by their average contribution magnitude in the surrogate model.

Feature	Mean absolute SHAP
debt_to_income_ratio	0.162630
monthly_income_ngn	0.133235
credit_history_months	0.105298
missed_payments_12m	0.088749
employment_years	0.030538
gender_Female	0.018006
gender_Male	0.012958
region_type_Urban	0.006979
region_type_Rural	0.006558
age	0.005531

Financial-history variables dominate the surrogate feature-importance ranking. The demographic indicator contributions are smaller in this ranking, but small global attribution does not prove absence of group effects or proxy effects. Feature importance does not establish causation, fairness, legal permissibility or the exact internal logic of a different model. A production assessment should include local explanations, stability checks, feature-dependence analysis and review of proxy variables.

6. Bias Mitigation Experiments

The project compared the original model with two separately constrained models: a gender-constrained model and a region-constrained model. The mitigation approach used Fairlearn ExponentiatedGradient with a DemographicParity constraint. The experiment demonstrates that optimising parity for one attribute can change outcomes for another attribute.

Evaluation	Original	Gender-constrained	Region-constrained
Test agreement with synthetic reference	0.87	0.94	0.94
Gender DP difference	0.1246	0.0162	0.0418
Gender selection-rate ratio	0.7243	0.9693	0.9235
Gender TPR difference	0.3106	0.0114	0.0265
Region DP difference	0.0944	0.1383	0.1271
Region selection-rate ratio	0.7850	0.7700	0.7888
Region TPR difference	0.0715	0.0014	0.0142

The gender-constrained experiment substantially reduced the gender selection-rate and TPR gaps, but the regional selection-rate difference increased. The region-constrained experiment improved the gender metrics relative to the original comparison and reduced the regional TPR difference; however, its regional selection-rate ratio remained below 0.80 and its regional DP difference was higher than the original. This illustrates that a single fairness constraint does not guarantee balanced outcomes across all groups.

The 0.80 selection-rate ratio sometimes used as a screening heuristic is not a universal legal test and should not be treated as proof of fairness or discrimination. Fairness definitions can conflict, and model choice must be justified by the system's purpose, applicable law, stakeholder impact and error costs. The 0.94 result is agreement with synthetic reference labels and must not be presented as real lending accuracy.

7. Intersectional Fairness Analysis

The intersectional audit combined gender and region into four groups. This helps identify patterns that may be hidden when each attribute is examined independently. The reported sample sizes differ by group and should be considered when interpreting rates.

Intersectional group	n	Original selection	Gender-mitigated	Region-mitigated
Female / Rural	70	0.3571	0.6000	0.6000
Female / Urban	95	0.3053	0.4737	0.5053
Male / Rural	53	0.5472	0.6038	0.6038
Male / Urban	82	0.3902	0.4512	0.4390

The mitigation experiments produced more closely clustered selection rates across these four synthetic groups. Nevertheless, residual differences remain, and the results are based on limited synthetic subgroup counts. Accuracy values also varied: original group accuracy ranged from 0.7571 to 1.0000; the gender-mitigated results ranged from 0.9245 to 0.9714; and the region-mitigated results ranged from 0.9268 to 0.9714. These figures require cautious interpretation because they are calculated against synthetic reference outcomes.

8. AI Governance and Risk Management

The project includes an AI risk register containing ten identified risks, with accountable owners and treatment information. The register records one Critical and nine High treatment-priority items. These are project-assigned priorities, not independently validated enterprise risk ratings.

The governance framework is structured around the four functions of the NIST AI Risk Management Framework: GOVERN, MAP, MEASURE and MANAGE. The framework is intended to connect organisational accountability, context identification, measurement and ongoing risk response.

Function	Application in this project
GOVERN	Assign accountability, define policies, document approvals, establish oversight and escalation routes.
MAP	Define the system context, intended purpose, affected stakeholders, data sources and potential harms.
MEASURE	Evaluate fairness, performance, explainability, robustness and relevant risks using documented methods.
MANAGE	Prioritise risks, apply mitigations, monitor outcomes, document residual risks and define stop/escalation criteria.

Recommended governance controls include a named model owner, independent validation before deployment, documented human review for adverse decisions, a route for applicants to challenge or correct data, change management, incident reporting, monitoring for drift and periodic fairness reassessment. The organisation should retain versioned datasets, code, model artefacts, test results and approval records to support auditability.

9. Legal and Regulatory Compliance Assessment

The repository contains a preliminary legal-compliance assessment and a 13-entry compliance matrix. The matrix references the Nigeria Data Protection Act 2023 (nine entries), the EU AI Act (two entries), UNESCO materials (one entry) and OECD AI principles (one entry). Its recorded statuses are six Pending, three Remediation required, two Conditional and two Recommended.

These entries are an initial issue-spotting framework, not a legal opinion or a certification of compliance. The existence of a law or framework in the matrix does not automatically establish that every provision applies to the fictional organisation. Applicability depends on facts such as establishment, processing activities, affected persons, territorial reach, system purpose and relevant statutory definitions. The EU AI Act should be assessed only after determining whether its territorial and material scope is engaged.

Before any real-world use, qualified counsel and the organisation's data-protection lead should verify the current authoritative texts, exact articles, commencement and transitional provisions, regulatory guidance, lawful basis, transparency duties, data-subject rights, DPIA requirements where applicable, processor/controller roles, retention, security, cross-border transfers and any sector-specific financial regulation. Any references in the existing matrix should be checked provision by provision and updated to the applicable facts and current law.

The project does not claim that SecureFinance Africa has violated a law, because the entity is fictional and the assessment does not establish the facts needed for such a conclusion. It also does not claim that the matrix's Pending or Remediation required statuses are final legal determinations.

10. Testing, Validation and Reproducibility

The project test suite contains six automated validation tests. The recorded execution result was six passed in approximately 3.43 seconds. Tests cover the presence and readability of the synthetic dataset, expected demographic groups, fairness output availability, intersectional group coverage and valid rates, governance/compliance documentation, and key figures.

The project uses a Python virtual environment. The recorded environment included Python 3.14.8 and pytest 9.2.0. The repository includes requirements files and a lock file to support dependency management. A clean working tree was confirmed after the initial GitHub push, and the main branch was configured to track the remote main branch.

A test pass confirms that the specified project checks succeeded; it does not prove that the analytical methods are free of defects, that the data represent real populations, or that a model is suitable for operational decisions. Reproducibility should be strengthened by preserving run logs, dependency versions, random seeds, input hashes and dated output manifests.

11. Limitations

• Synthetic data: results are not generalisable to real applicants or financial institutions.
• Synthetic reference labels: model agreement is not real-world predictive performance.
• Surrogate explainability: SHAP values describe the surrogate and are not causal explanations of the original system.
• Group metrics: aggregate rates can hide within-group variation; subgroup sizes and uncertainty matter.
• Fairness trade-offs: improving one metric or protected group comparison can worsen another.
• Legal scope: the compliance matrix is preliminary and has not been validated by legal counsel against the full facts and current law.
• No deployment validation: operational security, robustness, accessibility, monitoring, human factors and sector-specific controls are outside the demonstrated tests.

12. Recommendations and Action Plan

Priority	Recommended action	Evidence of completion
Immediate	Review and correct every legal-matrix citation and determine territorial/material applicability with qualified counsel.	Signed legal review; updated compliance matrix.
High	Conduct independent validation using lawful, representative, quality-assessed data and documented labels.	Validation report; data provenance and quality records.
High	Evaluate multiple fairness metrics across gender, region and intersectional groups, with uncertainty intervals and error-cost analysis.	Fairness assessment with subgroup counts and confidence intervals.
High	Establish human review, adverse-decision explanation, appeal and data-correction processes.	Approved operating procedure and user-facing notices.
High	Approve model governance roles, risk acceptance and deployment gates.	Named owners; signed approvals; residual-risk register.
Ongoing	Monitor drift, outcome disparities, incidents, complaints and model changes; define rollback triggers.	Monitoring dashboard, review minutes and incident log.

13. Conclusion

The AICS201 project demonstrates an integrated assessment workflow for an illustrative AI-enabled financial decision system. It generated a synthetic dataset, measured fairness differences, explored surrogate-model explanations, tested fairness-constrained mitigation, examined intersectional outcomes, documented AI risks and created a preliminary compliance matrix and governance framework.

The results show why AI ethics and governance require more than a single accuracy score. The initial synthetic audit identified group disparities; mitigation experiments reduced some gaps while worsening or retaining others; and the intersectional analysis highlighted the value of examining combined demographic groups. Explainability outputs provide a useful starting point for review but do not establish causality or fairness.

The work should therefore be treated as an academic prototype and assessment exercise. It is not evidence that a real system is compliant, fair, safe or ready for deployment. Any operational use would require representative data, independent validation, legal review, meaningful human oversight, transparent user processes and continuous monitoring.

Appendix A. Key Metrics Summary

Metric	Reported value	Interpretation
Initial audit accuracy	0.9025	Overall accuracy on the synthetic audit evaluation.
Initial gender DP difference	0.1242	Female/male selection-rate gap.
Initial gender EOpp difference	0.2511	Female/male true-positive-rate gap.
Initial region DP difference	0.0665	Rural/urban selection-rate gap.
Initial region EOpp difference	0.1639	Rural/urban true-positive-rate gap.
Surrogate test accuracy	0.9033	Agreement of surrogate with target outputs on its test set.
Mitigation model agreement	0.94	Agreement with synthetic reference labels; not real lending accuracy.

Appendix B. Project Files and Outputs

File / folder	Purpose
README.md	Project overview and usage notes.
data/synthetic/	Synthetic input dataset.
scripts/	Dataset-generation script.
fairness_analysis/	Fairness, mitigation and intersectional analysis scripts.
explainability/	SHAP surrogate-analysis script.
risk_assessment/	AI risk register.
legal_compliance/	Preliminary legal assessment and compliance matrix.
docs/	Data dictionary and governance framework.
outputs/reports/	CSV and text analytical outputs.
outputs/figures/	Fairness and explainability figures.
tests/	Automated project-validation tests.
evidence/screenshots/	Location intended for genuine project evidence screenshots.

Evidence note: insert or append genuine screenshots captured from the actual project execution if required by the assessment brief. The report does not claim that screenshots are included in this PDF.

End of report